# Natural Language Generation (NLG)

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# Outline

- **Introduction to NLG**
- **NLG Artchitecture**
  - **1. Architectures by Design Philosophy**
  - **2. Architectures by Underlying Technology**
  - **3. Extractive vs. Abstractive (Output Style)**

# NATURAL LANGUAGE GENERATION

- **Definition:** A subfield of AI and NLP that automatically converts data into natural language.
  - Natural Language Generation is the process of converting **structured or unstructured data** into coherent and contextually meaningful **natural language text.**
- **Key Concept:** Think of NLP as the entire toolkit; **NLG is the specialized tool** used for **writing**.
- **Core Goal:** To bridge the gap between raw data and human understanding.

# NATURAL LANGUAGE GENERATION

NLG = Data → Text

Input: Numbers / Data / Facts

Output: Human-readable sentences

Example: Temperature = 35°C → 'It is very hot today.'

# NATURAL LANGUAGE GENERATION

Key Characteristics

- Produces human-like sentences
- Uses grammar and context
- Generates sentences, paragraphs, reports

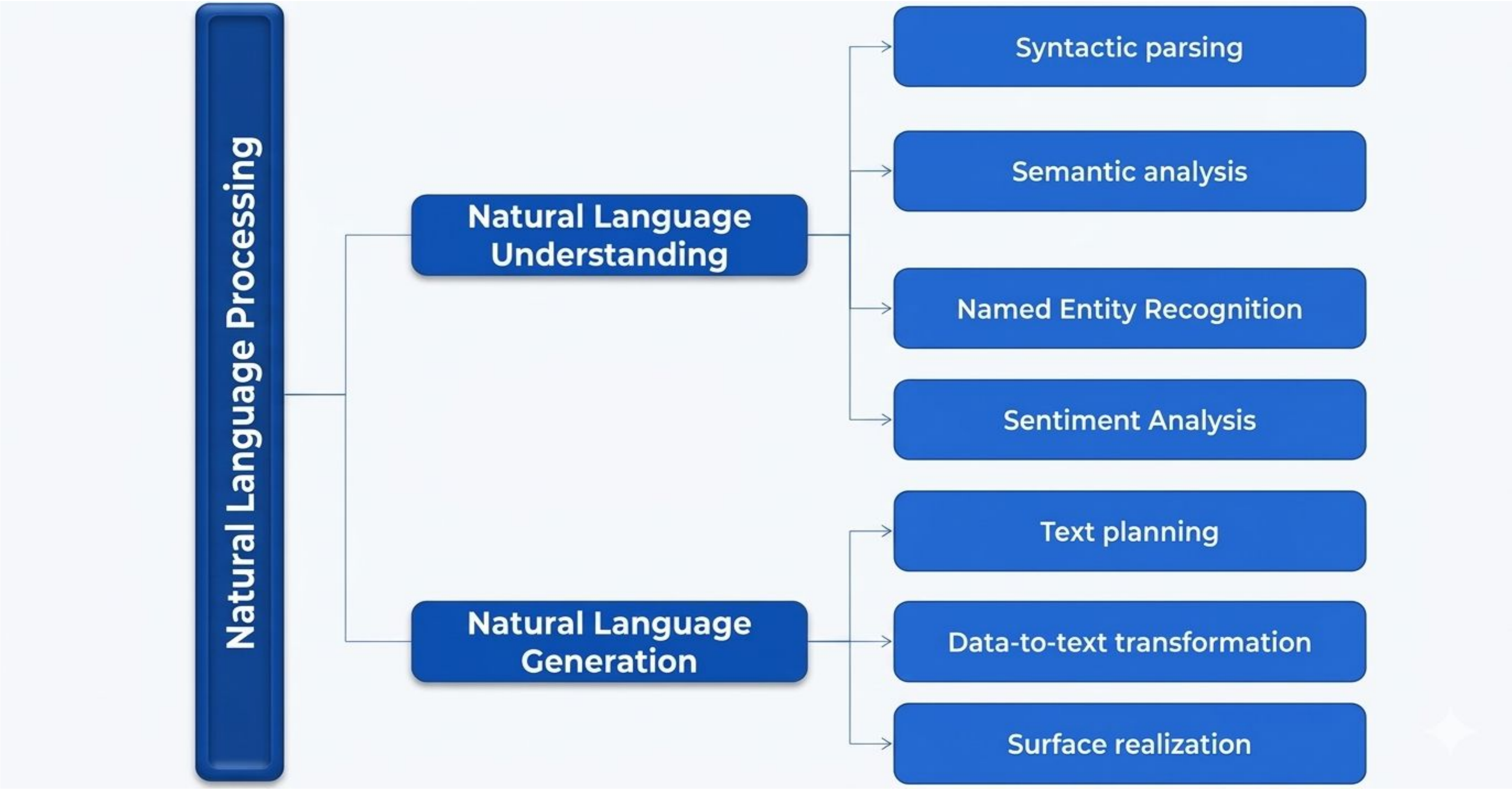
# NLG –Data vs Text

| Feature | Data | Text |
|---|---|---|
| **Primary Audience** | Machines/Algorithms | Humans |
| **Organization** | Rigid (Tables/Code) | Fluid (Sentences) |
| **Complexity** | Easy for computers to calculate | Easy for humans to understand |
| **Role in NLG** | The "Raw Material" | The "Finished Product" |

# NATURAL LANGUAGE GENERATION

## NLG vs. NLU vs. NLP

| Feature | NLP | NLU | NLG |
|---|---|---|---|
| Primary Role | Broad Umbrella | The “Reader” | The “Writer” |
| Function | Processing Language | Understanding intent | Generating text |
| Input Type | Text/Speech | Human Language | Structures Data |


Natural Language Processing
Natural Language Understanding
Natural Language Generation
Syntactic parsing
Semantic analysis
Named Entity Recognition
Sentiment Analysis
Text planning
Data-to-text transformation
Surface realization

# NATURAL LANGUAGE GENERATION

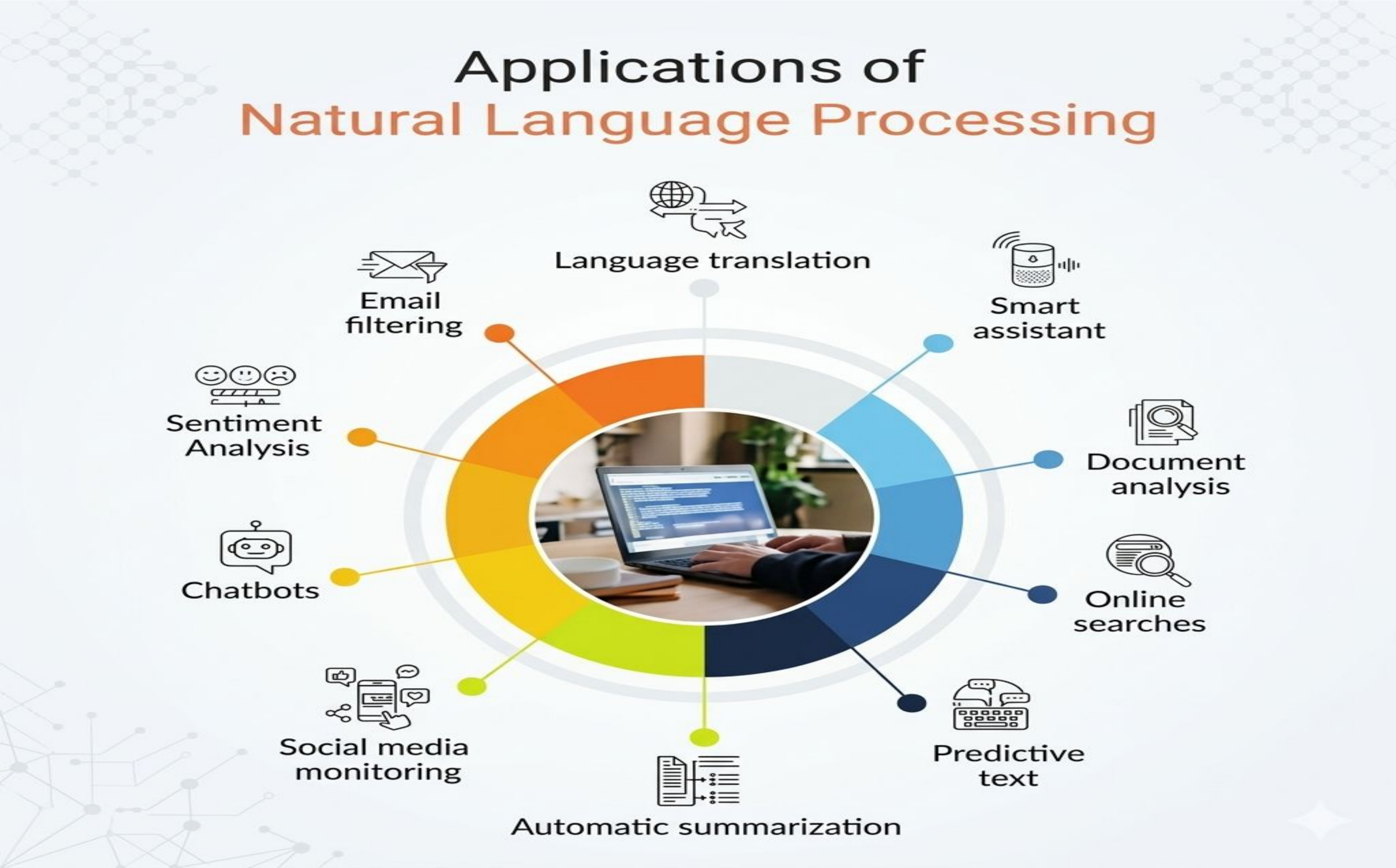
## Why NLG is Important

- Saves time in report writing
- Handles large data efficiently
- Enables human-computer interaction
- Widely used in modern AI systems

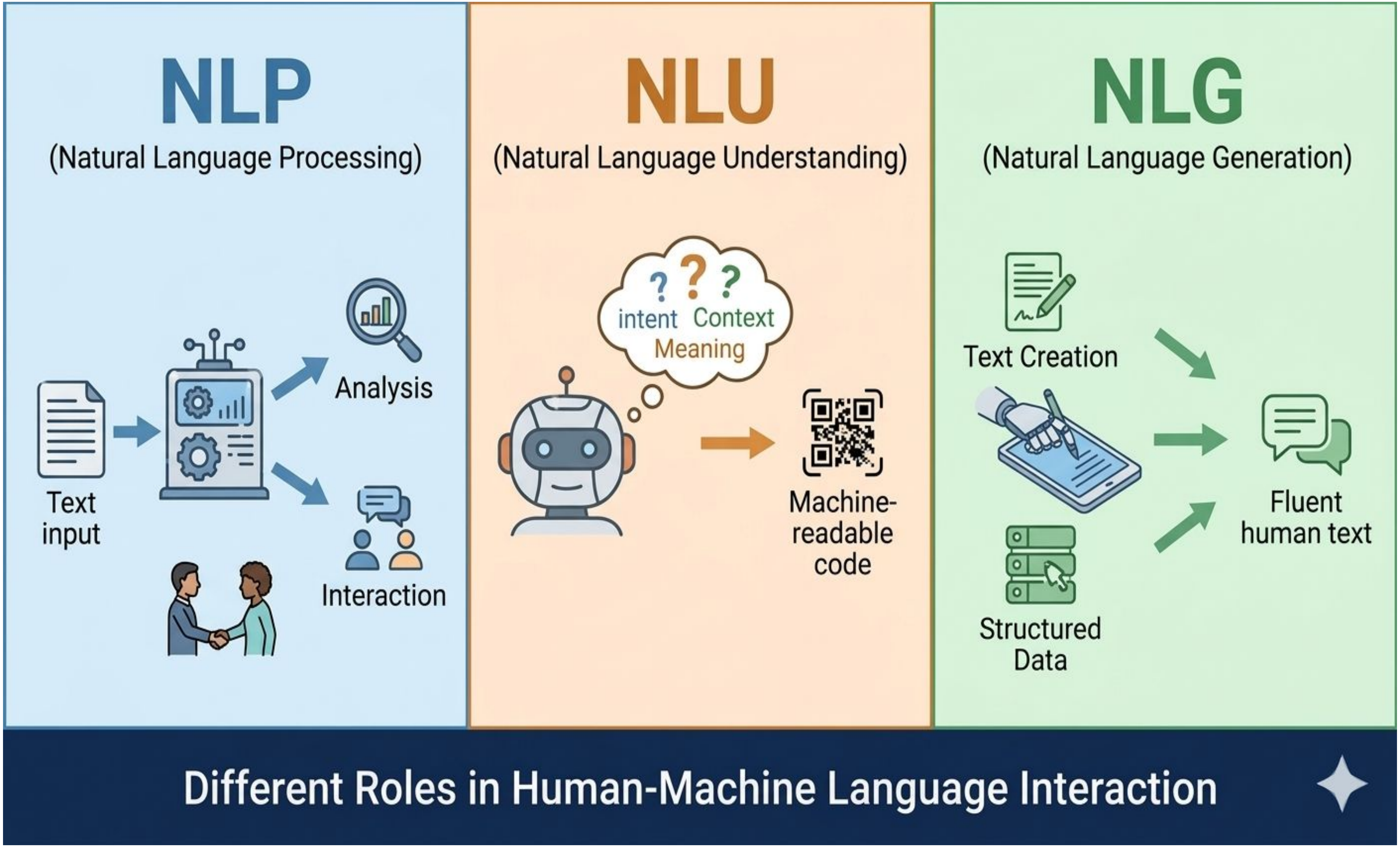
# NATURAL LANGUAGE GENERATION

## Real-World Applications

- **Conversational AI:** Powering chatbots and voice assistants like Alexa/Siri.
- **Automated Journalism:** Creating quick reports for weather, sports, and finance.
- **Business Intelligence:** Turning complex spreadsheets into written summaries.
- **Personalization:** Generating custom product descriptions and marketing emails.



Applications of
Natural Language Processing
Language translation
Email filtering
Smart assistant
Sentiment Analysis
Document analysis
Chatbots
Online searches
Social media monitoring
Predictive text
Automatic summarization



NLP
(Natural Language Processing)
Text input
Analysis
Interaction
NLU
(Natural Language Understanding)
? ? ?
intent
Context
Meaning
Machine-readable code
NLG
(Natural Language Generation)
Text Creation
Structured Data
Fluent human text
Different Roles in Human-Machine Language Interaction

## NLG Architecture

NLG architecture broadly categorized based on its **design philosophy** (process information) and its **underlying technology** (the algorithms used)

### 1. Architectures by Design Philosophy

This categorization describes the structural flow of information through the system.

# NATURAL LANGUAGE GENERATION

**a)Pipeline Architecture (The Classic Standard):** The most common modular approach, where tasks are performed in a strict sequence:

**Macro-planning->Micro-planning->SurfaceRealization**.

**b)Interleaved Architecture:** Steps in the generation process occur simultaneously or overlap, allowing later stages (like word choice) to influence earlier ones (like content selection).

**c)Integrated/End-to-End Architecture:** A modern "one-step" approach where a single model (typically a neural network) maps input data directly to output text, bypassing distinct intermediate modules.

# NATURAL LANGUAGE GENERATION

**2. Architectures by Underlying Technology**

These types define the "intelligence" level and technical implementation of the generator.

The architectures are categorized in to

- Template-Based
- Rule Based
- Statistical
- Neural

# NATURAL LANGUAGE GENERATION

| Architecture | Description | Key Characteristic | Example |
|---|---|---|---|
| **Template-Based** | Uses fixed text frames with "blanks" for data variables. | High precision, low variety. | Simple weather alerts, sports scores. |
| **Rule-Based** | Relies on complex linguistic rules and "if-then" logic to build sentences. | Highly controllable and predictable. | SimpleNLG, RealPro. |
| **Statistical** | Uses probability models trained on large text corpora to predict word sequences. | More natural than rules; requires lots of data. | Early versions of Google Translate. |
| **Neural (Modern)** | Uses deep learning (Transformers, RNNs) to learn language patterns directly. | Highly fluent and context-aware. | **GPT-4**, T5, Claude. |

# NATURAL LANGUAGE GENERATION

## 3. Extractive vs. Abstractive (Output Style)

While these are often called "types" of NLG, they actually represent different architectural goals.

- **Extractive:** The architecture acts like a **highlighter**, pulling existing phrases directly from source data to create a summary.
  Example: legal/technical summaries

- **Abstractive:** The architecture acts like a **writer**, generating entirely new sentences by "understanding" the core concepts.
  Example: Creative contents (ChatGPT)

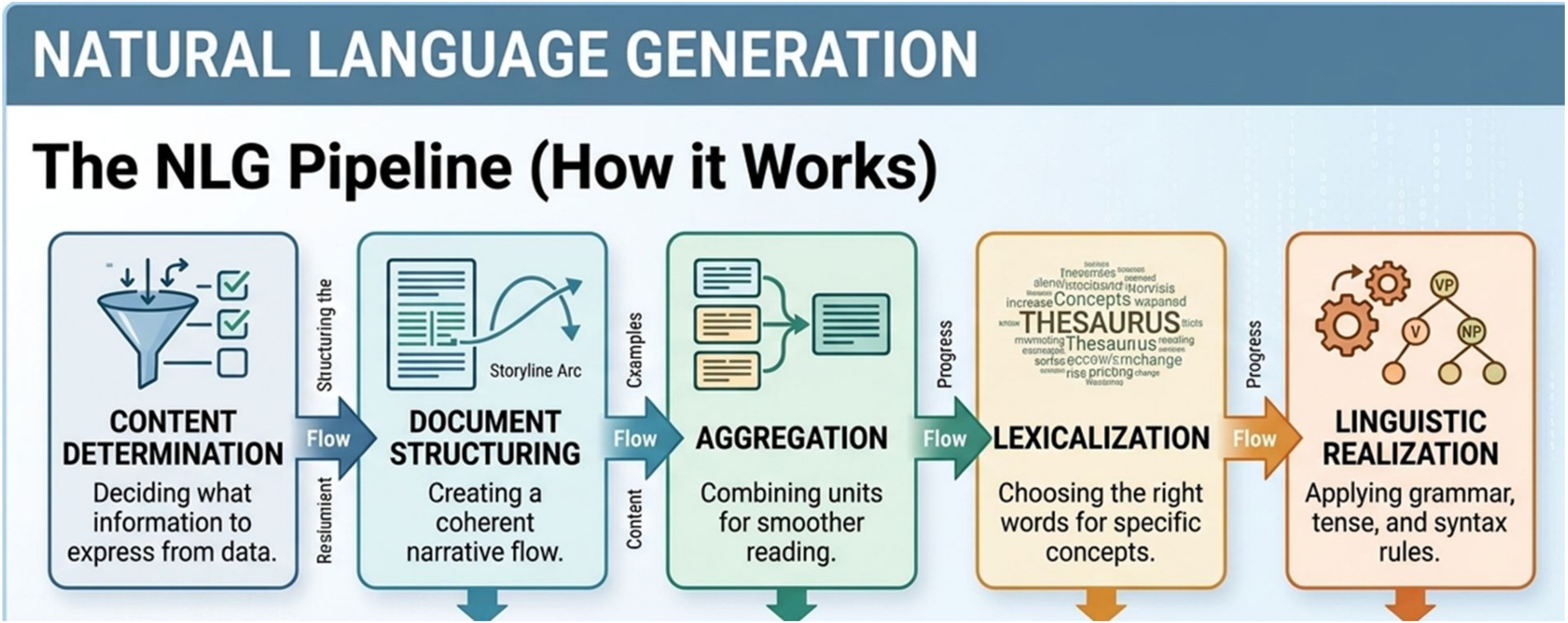
# NATURAL LANGUAGE GENERATION

## 1. Architectures by Design Philosophy

### a)The NLG Pipeline (How it Works)

- **Content Determination:** Deciding what information to express.
- **Document Structuring:** Creating a coherent narrative flow.
- **Aggregation:** Combining units for smoother reading.
- **Lexicalization:** Choosing the right words for specific concepts.
- **Linguistic Realization:** Applying grammar, tense, and syntax rules

# NATURAL LANGUAGE GENERATION



# NATURAL LANGUAGE GENERATION

**NLG Pipeline** (real-world example ) :

**Generating a Weather Report:**

Imagine the raw data is: Temp: 39°C, Condition: Sunny, Location: Vijayawada, Time: 2:00 PM.

## 1. Content Determination

- **What it is:** Deciding *which* specific data points are important enough to mention. We can’t include everything, or the text becomes cluttered.

- **Example:** Out of all the sensor data (humidity, wind speed, pressure, Temperature etc.), the system decides that **Temperature** and **Sunny Condition** are the most relevant for a quick update. It ignores the barometric pressure because a regular user doesn't need it.

## 2. Document Structuring

- **What it is:** Planning the "story arc." It decides the order in which the chosen facts will be presented so they make sense to a human.

- **Example:** The system decides to start with the **Location** first, then the **Current Weather**, and finally a **Recommendation**.

    - *Plan:* [Location] -> [Condition/Temp] -> [Advice].

## 3. Aggregation

- **What it is:** Merging related facts into a single sentence to avoid sounding like a repetitive robot.

- **Example:** Instead of saying *"It is 32°C. It is sunny,"* the system aggregates them into: *"It is a sunny 32°C day."* This makes the flow feel more natural.

## 4. Lexicalization

- **What it is:** Choosing the actual "human" words or adjectives to describe the data. This sets the **tone**.

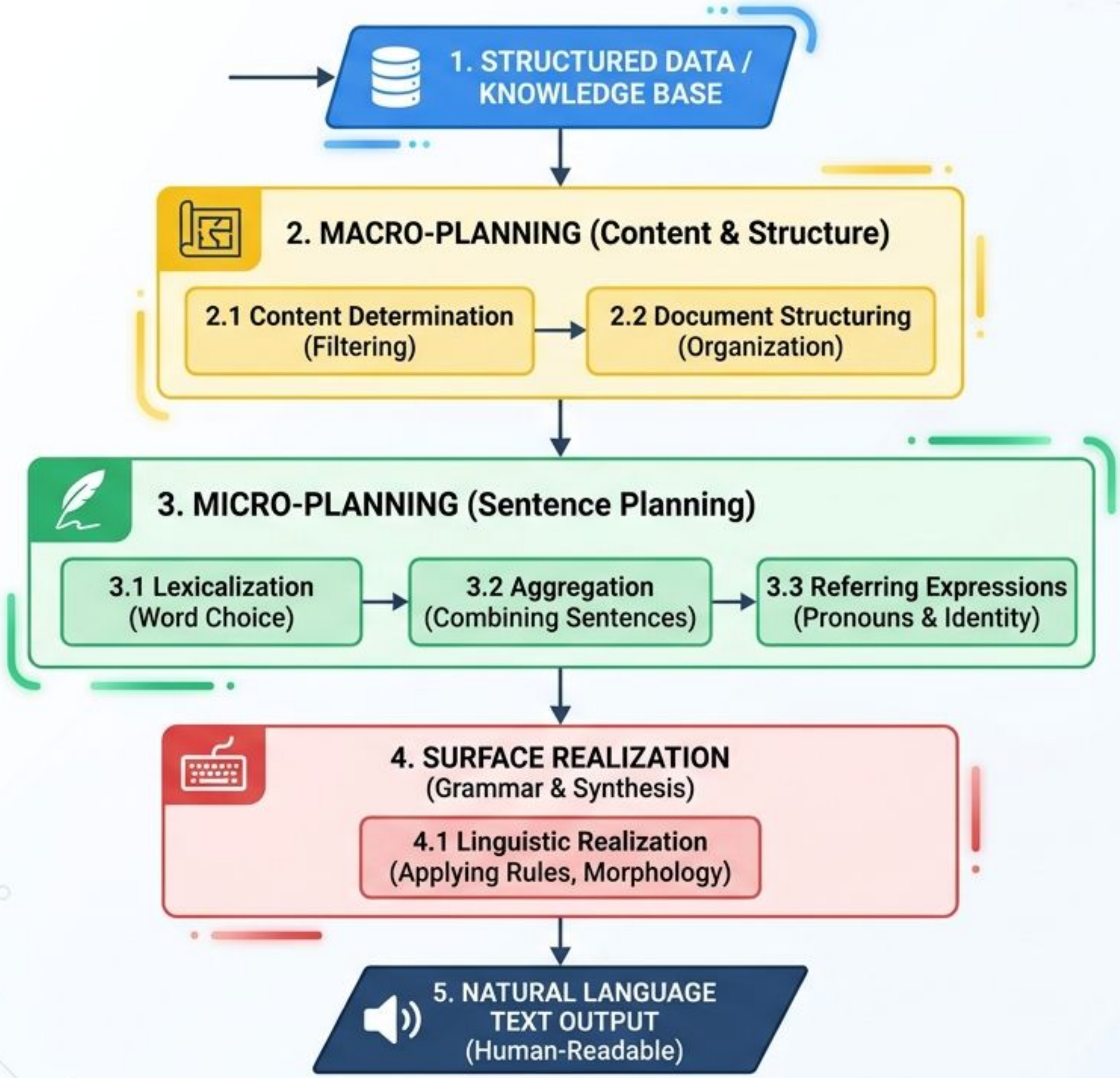
- **Example:** Since the data is 32°C, the system chooses the word **"sweltering"** or **"hot"** rather than just "warm." It turns the raw number into a descriptive word.

# NATURAL LANGUAGE GENERATION

## 5. Linguistic Realization

- **What it is:** The final "polish." The system applies rules of grammar, punctuation, and tense to create the final string of text.

- **Example:** It adds "The," "is," and proper punctuation to create the final output: "It *is currently a sweltering 32°C in Vijayawada; don't forget your sunscreen!"*


1. STRUCTURED DATA / KNOWLEDGE BASE
2. MACRO-PLANNING (Content & Structure)
2.1 Content Determination (Filtering)
2.2 Document Structuring (Organization)
3. MICRO-PLANNING (Sentence Planning)
3.1 Lexicalization (Word Choice)
3.2 Aggregation (Combining Sentences)
3.3 Referring Expressions (Pronouns & Identity)
4. SURFACE REALIZATION (Grammar & Synthesis)
4.1 Linguistic Realization (Applying Rules, Morphology)
5. NATURAL LANGUAGE TEXT OUTPUT (Human-Readable)

# NATURAL LANGUAGE GENERATION

## Pros (Advantages)

- **Modularity & Focus:** Because the system is broken into clear stages, developers can work on the "Writer" (Micro-planner) without touching the "Editor" (Macro-planner).
- **Explainability:** If the output is wrong, it is easy to trace the error. You can see exactly which stage failed (e.g., "The data was selected correctly, but the grammar was wrong").
- **Maintainability:** You can swap out one part of the pipeline. (eg: replace an English "Surface Realizer" with a French one to support a new language without changing your core data logic.
- **Stability:** It is highly predictable. Unlike AI models that might "hallucinate," a pipeline strictly follows the data it is given.

# NATURAL LANGUAGE GENERATION

**Cons (Disadvantages)**

- **The "Pipeline Slump" (Error Propagation):** If the first stage makes a mistake (like picking the wrong data), the later stages can't fix it. They just "polish" the mistake.
- **Lack of Feedback:** Information only flows forward. The Macro-planner doesn't know if the Micro-planner will struggle to find a good word for the data it just picked.
- **Inflexibility to Constraints:** As we saw in the "Interleaved" example, a pipeline can't easily handle strict character limits or specific tones because the content is "locked in" too early.
- **Computational Overhead:** Passing complex data structures between multiple modules can sometimes be slower and more complex to manage than a single "end-to-end" model.

## b) Interleaved Architecture

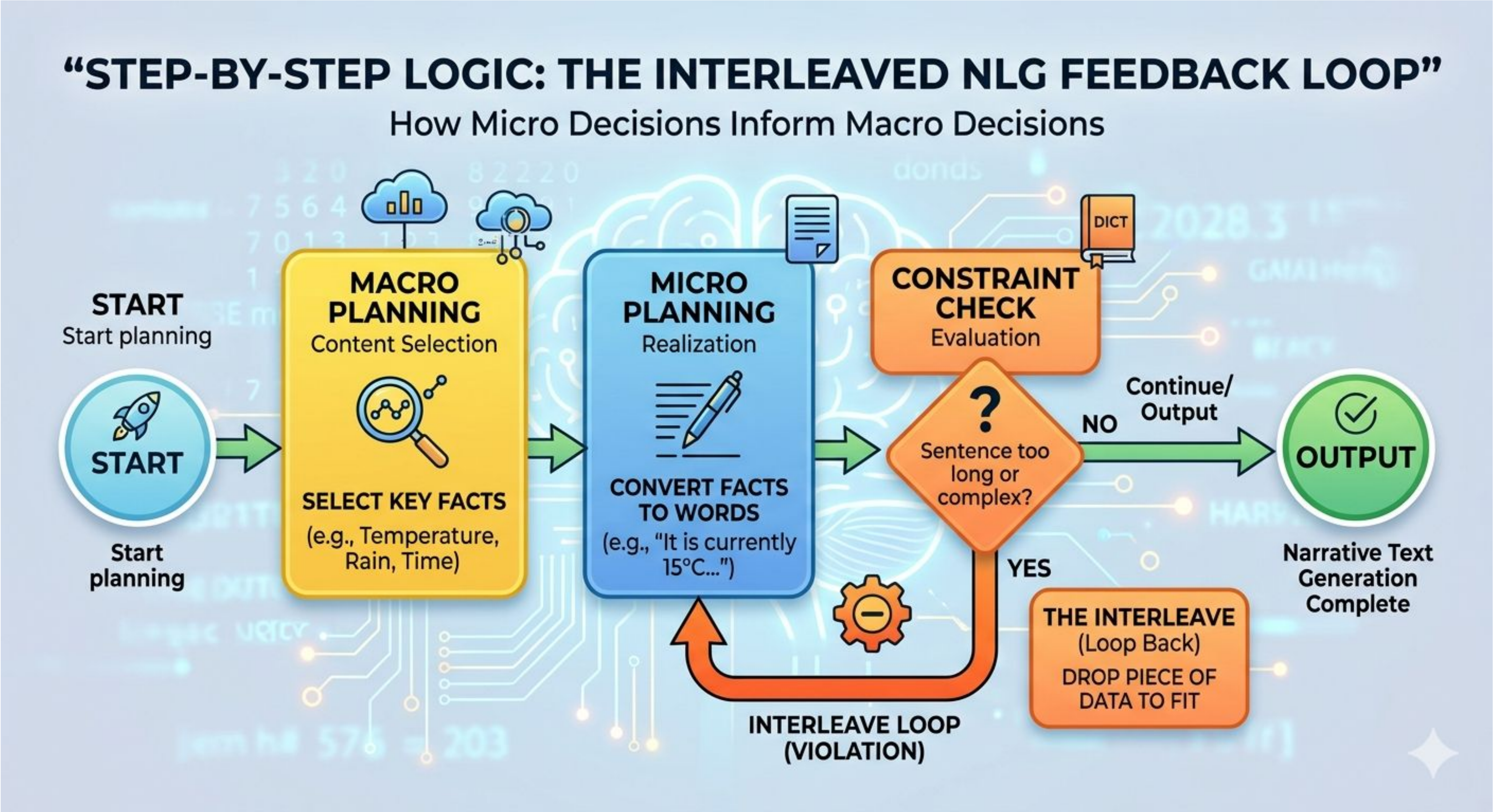
The stages of generation (Macro and Micro planning) are not a one-way street. Instead, they "talk" to each other in real-time.

A common reason for this is **Constraints** (like a character limit). If the system realizes it is running out of space while writing a sentence, it can "loop back" and change what it originally planned to say.

# NATURAL LANGUAGE GENERATION

## Step-by-Step Logic (Interleaved Architecture)

1. **Start Planning:** The system picks the most important facts (Macro).
2. **Start Writing:** It begins converting facts into words (Micro).
3. **Check Constraint:** It checks if the sentence is becoming too long or complex.
4. **The Interleave (Loop Back):** If the constraint is violated, the system **drops** a piece of data it previously selected to make the sentence fit.



"STEP-BY-STEP LOGIC: THE INTERLEAVED NLG FEEDBACK LOOP"
How Micro Decisions Inform Macro Decisions
START
Start planning
START
Start planning
MACRO PLANNING
Content Selection
SELECT KEY FACTS
(e.g., Temperature, Rain, Time)
MICRO PLANNING
Realization
CONVERT FACTS TO WORDS
(e.g., "It is currently 15°C...")
CONSTRAINT CHECK
Evaluation
DICT
?
Sentence too long or complex?
NO
Continue/ Output
YES
OUTPUT
Narrative Text Generation Complete
THE INTERLEAVE
(Loop Back)
DROP PIECE OF DATA TO FIT
INTERLEAVE LOOP
(VIOLATION)

# NATURAL LANGUAGE GENERATION

**NLG Interleaved Architecture** (real-world example ) :

**Automated Smart Watch Notification:**

The watch has a very small screen (a **Spatial Constraint**), so the "Micro" decision (how much space is left) must control the "Macro" decision (what data to include).

**NLG Interleaved Architecture** (real-world example ) :

**Automated Smart Watch Notification:**

**Step-by-Step Logic in Action**

1. **Macro-planning (The Initial Plan):**
   The system decides to include everything: *"Meeting with CEO at 9:00 AM. Traffic is heavy. Leave now to arrive on time."*
2. **Micro-planning (The Writing):**
   The system begins realizing the text into words:
   *"Meeting with CEO at 9:00 AM..."*

# **NLG Interleaved Architecture** (real-world example ) :

# Automated Smart Watch Notification:

**3. Constraint Check (The Evaluation):**
The system "measures" the pixels. **Result:** *"This is already 3 lines long. The watch face only fits 2 lines!"*
**4. The Interleave (The Loop Back):**
The Micro-planner sends a "Space Alert" back to the Macro-planner. The Macro-planner **drops** the "CEO" and "9:00 AM" details because the user already knows who they are meeting.
**5. New Output (The Success):**
*"Heavy traffic! Leave now to arrive on time."* (**Fits perfectly on the screen.**)

**c) Integrated or End-to-End (E2E) Architecture** represents a "one-step" approach to Natural Language Generation. Unlike the classic pipeline where you have an Architect (Macro) and a Writer (Micro), an E2E system uses a single, massive neural network to go directly from **Data** to **Text**.

# NATURAL LANGUAGE GENERATION

**1. The Core Concept: No More Hand-Offs**

In a pipeline, you have to manually code the rules for how data becomes a "plan" and how a "plan" becomes a "sentence." In an **Integrated** system:

- **The Model:** Usually a Transformer-based Large Language Model (like **GPT-4**, **T5**, or **Llama**).
- **The Process:** The data is fed into the model as a prompt, and the model "predicts" the entire response in one go.
- **The Learning:** The system learns how to structure, select, and write by looking at millions of examples of data-text pairs (Training).

## 2. A Clear Example: The "Restaurant Review" Generator

**Input Data:**

{ "Name": "The Pizza Place", "Rating": 5, "Food": "Italian", "Price": "Cheap" }

**The Pipeline Approach (Modular):**

1. **Macro:** Decides to mention Name, Food, and Price.
2. **Micro:** Chooses the word "affordable" for "cheap."
3. **Realizer:** Builds:"The Pizza Place is an affordable Italian spot."

# NATURAL LANGUAGE GENERATION

**The Integrated/E2E Approach (Neural):**

- The model receives the JSON and immediately generates:
  *"If you're looking for great Italian on a budget,*
  *The Pizza Place is a 5-star gem you can't miss!"*
- **What happened?** The model did the selecting, ordering, and writing **simultaneously** inside its neural layers.

# NATURAL LANGUAGE GENERATION

## 3. Pros and Cons

| Feature | Integrated / End-to-End |
| --- | --- |
| Pros (The Good) | **Human-like Fluency:** Produces much more natural, varied, and creative text.<br>**Low Manual Effort:** No need to write thousands of "if-then" grammar rules. |
| Cons (The Bad) | **Hallucinations:** The model might "lie" and say the food is "Spicy" even if the data doesn't say so.<br>**Black Box:** It's hard to explain *why* the model chose a specific word. |

# 2. Architectures by Underlying Technology

## a) Template-Based NLG

**Template-Based NLG** is the simplest and most common form of natural language generation. It uses a "fill-in-the-blanks" approach where a fixed text string (the template) contains predefined slots for dynamic data variables.

# Template-Based NLG (continue..)

## 1. How it Works

The system maps structured data points directly into specific placeholders within a sentence. There is no complex linguistic "reasoning"; the output is always predictable based on the template used.

- **The Template:** "The temperature in [city] is [temp] degrees."
- **The Data:** { "city": "Mumbai", "temp": 32 }
- **The Output:** "The temperature in Mumbai is 32 degrees."

## 2. Key Characteristics

- **High Precision:** Because the sentences are pre-written by humans, the grammar is always perfect and the tone is exactly what the designer intended.
- **Low Complexity:** It does not require a "Macro" or "Micro" planning pipeline. It only requires a mapping function to plug data into slots.
- **Low Variety:** The system sounds repetitive. It will say the exact same sentence every time unless you manually write multiple templates for the same event.

# 3. Pros and Cons

| Pros (Advantages) | Cons (Disadvantages) |
|---|---|
| **Speed:** Extremely fast and computationally "cheap." | **Repetitive:** Users quickly notice the "robotic" pattern. |
| **Reliability:** 0% chance of "hallucinating" or lying. | **Maintenance:** Scaling to thousands of products requires writing thousands of templates. |
| **Control:** Ideal for legal or financial reports where wording must be exact. | **No Context:** Cannot handle complex linguistic changes (like singular vs. plural) automatically. |

# NATURAL LANGUAGE GENERATION

## 4. Real-World Applications

- **Banking:** "Your account ending in XXXX has a balance of $Y."
- **E-commerce:** Shipping updates and order confirmations.
- **Weather Apps:** Simple "Current condition in City is Sunny."

# NATURAL LANGUAGE GENERATION

## b) Rule-Based NLG

**Rule-Based NLG** (like the famous **SimpleNLG** library) acts more like a **Digital Grammarian**. It doesn't just fill in blanks; it uses linguistic "if-then" logic to build sentences that change based on the data.

# NATURAL LANGUAGE GENERATION

## 1. How it Works: The Logic Engine

Instead of a fixed string, the system stores **Linguistic Rules**. It maps data to grammatical roles (Subject, Verb, Object) and applies rules for:

- **Morphology:** Changing "run" to "runs" or "ran" automatically.
- **Agreement:** Ensuring "The **apple is**" vs. "The **apples are**."
- **Pronominalization:** Replacing "John" with "He" in the second sentence.

# Comparison: Template vs. Rule-Based

| Feature | Template-Based | Rule-Based (SimpleNLG Style) |
|---|---|---|
| Logic | "Data A goes in Slot B." | "If Data A is > X, use Word Y." |
| Grammar | Fixed/Manual. | Calculated/Dynamic. |
| Variety | Low (Sounds like a robot). | Higher (Sounds more natural). |
| Effort | Easy to set up. | Harder (Needs a linguist to write rules). |

## 3. Why use Rule-Based over Templates?

- **Scalability:** One rule for "Pluralization" works for 1,000 different words. With templates, you'd have to write two versions for every single sentence.

- **Multilingual Support:** You can keep the "Logic" (Macro-planning) the same and just swap the "Grammar Rules“ (Surface Realization) to translate the report into another language.

# NATURAL LANGUAGE GENERATION

## 1. Statistical NLG (The Probability Model)

Before the AI explosion, this was the standard. It uses **probabilities** to decide which word should follow another.

- **The Logic:** If the system sees the word "New," it calculates that the next word is likely "York" (80%) or "Delhi" (15%).
- **The Problem:** It has a "short memory." It might start a sentence well but forget the beginning by the end, leading to nonsensical results.

# NATURAL LANGUAGE GENERATION

## 2. Neural NLG (The Transformer/LLM)

This technology powers GPT-4, Claude, and AI on Google Search, which is powered by the Gemini family of models. It uses **Deep Learning** to understand the deep relationships between words across long distances.

- **The Logic:** Words are converted into numbers (vectors) in a high-dimensional space. The system understands the *concept* of a city, a location, and a context, not just that "York" follows "New."
- **The Process:** It takes **Structured Data** as a prompt and generates a natural response based on its training.

# Template-Based vs Rule-Based vs Neural

| Feature | Template-Based | Rule-Based | Neural (E2E) |
|---|---|---|---|
| Analogy | A Form Letter | A Grammarian | A Creative Writer |
| Logic | Fill-in-the-blanks | Manual "If-Then" rules | Neural Probabilities |
| Accuracy | 100% (Fixed) | High (But complex) | Variable (Hallucinations) |
| Fluency | Low (Robotic) | Moderate | High (Human-like) |
| Best For | Banking alerts | Technical manuals | Chatbots & Stories |

# NATURAL LANGUAGE GENERATION

******ANY Queries******

# Modern NLG Models

- • GPT Models
- • Prompt Engineering
- • Fine-tuning Concepts

# Machine Translation Overview

- • Definition
- • Importance
- • Applications

# Problems in Machine Translation

- • Ambiguity
- • Idioms
- • Morphology
- • Context issues

# Indian Language Challenges

- • Rich Morphology
- • Free Word Order
- • Script Diversity

# MT Approaches

- • Rule-Based MT
- • Statistical MT
- • Neural MT

# Neural Machine Translation

- • Encoder-Decoder
- • Attention Mechanism
- • Transformers

# Indian Language Translation

- • Hindi-English
- • Telugu-English
- • Government Initiatives

# Introduction to NLG

- Definition of NLG
- NLG vs NLP vs NLU
- Applications (ChatGPT, Reports, Chatbots)

# What is NLG?

- Branch of Artificial Intelligence
- Automatically generates human-like language
- Converts data into meaningful text or speech

# Simple Understanding

- NLG = Data → Text
- Input: Numbers / Data / Facts
- Output: Human-readable sentences
- Example: Temperature = 35°C → 'It is very hot today.'

# Formal Definition

- Natural Language Generation is the process of converting structured or unstructured data
- into coherent and contextually meaningful natural language text.

# Key Characteristics

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# NLG vs NLU

- NLG: Generates text (Data → Language)
- NLU: Understands text (Language → Meaning)

# Real-World Applications

- Chatbots
- Weather reports
- Financial report generation
- Voice assistants
- Automated news writing

# Why NLG is Important

- Saves time in report writing
- Handles large data efficiently
- Enables human-computer interaction
- Widely used in modern AI systems

# One-Line Summary

- NLG enables machines to communicate like humans
- by converting data into natural language.

# Quick Recap

- NLG = Data → Language
- Part of AI
- Opposite of NLU
- Used in real-world systems

# NLP vs NLU vs NLG

# What is NLP?

- Natural Language Processing
- Overall AI field dealing with human language
- Includes understanding and generation

# What is NLU?

- Subfield of NLP
- Focuses on understanding language
- Extracts meaning from text/speech
- Language → Meaning

# NLU Example

- Input: Book a flight to Delhi tomorrow
- Intent: Booking
- Location: Delhi
- Time: Tomorrow

# What is NLG?

- Subfield of NLP
- Focuses on generating language
- Converts data into sentences
- Data → Language

# NLG Example

- Input: Weather data
- Output: It will rain today

# Relationship

- NLP = NLU + NLG
- NLU understands input
- NLG generates output

# Comparison

- NLP: Complete system
- NLU: Understanding
- NLG: Generation

# Real-Life Chatbot Flow

- User asks question
- NLU understands
- System processes
- NLG generates response

# Summary

- NLP = Whole system
- NLU = Understand
- NLG = Generate

# Architecture of NLG Systems

- • Content Determination
- • Document Structuring
- • Lexicalization
- • Surface Realization

# Pipeline vs End-to-End NLG

- • Rule-based systems
- • Machine Learning models
- • Neural NLG approaches

# Generation Tasks

- • Data-to-text
- • Text Summarization
- • Dialogue Generation
- • Story Generation

# Representations in NLG

- • Meaning Representation
- • Semantic Frames
- • Knowledge Graphs

# Applications of NLG

- • Healthcare
- • Finance
- • Chatbots
- • Smart Assistants

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- • Telugu-English
- • Government Initiatives

# Natural Language Generation & Machine Translation

Guest Lecture (10 Hours)

Prepared by: Dr. Jagadeesh M S

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- NLG vs NLP vs NLU
- Applications (ChatGPT, Reports, Chatbots)

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- • Machine Learning models
- • Neural NLG approaches

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- • Meaning Representation
- • Semantic Frames
- • Knowledge Graphs

# Applications of NLG

- • Healthcare
- • Finance
- • Chatbots
- • Smart Assistants

# Modern NLG Models

- • GPT Models
- • Prompt Engineering
- • Fine-tuning Concepts

# Machine Translation Overview

- • Definition
- • Importance
- • Applications

# Problems in Machine Translation

- • Ambiguity
- • Idioms
- • Morphology
- • Context issues

# Indian Language Challenges

- • Rich Morphology
- • Free Word Order
- • Script Diversity

# MT Approaches

- • Rule-Based MT
- • Statistical MT
- • Neural MT

# Neural Machine Translation

- • Encoder-Decoder
- • Attention Mechanism
- • Transformers

# Indian Language Translation

- • Hindi-English
- • Telugu-English
- • Government Initiatives

# NATURAL LANGUAGE GENERATION

- **Definition:** A subfield of AI and NLP that automatically converts data into natural language.

- **Key Concept:** If NLP is the broad umbrella, NLG is the **"Writer"**.

- **Core Goal:** To bridge the gap between raw data and human understanding.

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# NATURAL LANGUAGE GENERATION

## Key Types of NLG

- **Extractive NLG:**
  - Pulls exact sentences/phrases directly from source text.
  - Best for high-precision summaries (legal/technical).
- **Abstractive NLG:**
  - Generates entirely new text based on understood concepts.
  - Produces more creative and human-like output (GPT-4).